

Trainings ▾

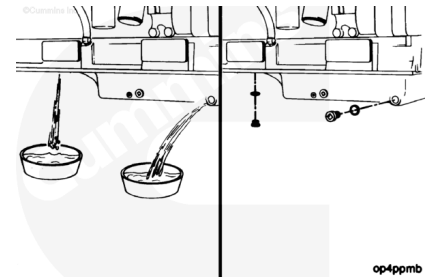
Drain

⚠ WARNING ⚠

To reduce the possibility of personal damage, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



Note : Drain the engine oil when the temperature is approximately 60°C [140°F].

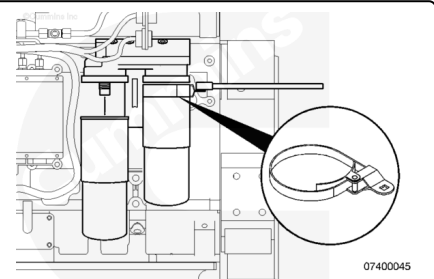
Remove the oil drain plug and copper washers from the bottom of the oil sump and the oil pan adapter cover plate and drain the oil.

Remove

Note : The following steps show the combination oil filter. Use the same procedures when changing the remote bypass oil filters.

Remove the oil filters with oil filter wrench, Part Number 3400157 or 3400158, or equivalent.

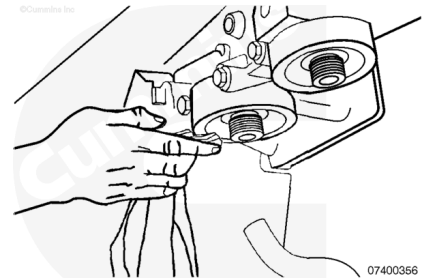
Discard the oil filters if they are **not** needed for failure analysis.



Install

Clean the oil filter head surface.

Lubricate the gasket surface of the oil filter with clean engine oil.



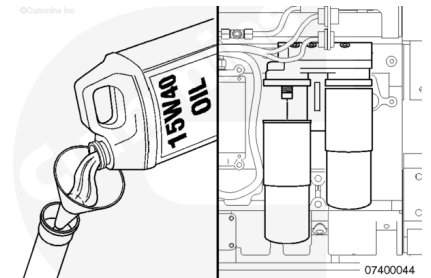
Fill the oil filters with clean 15W-40 oil.

Install the oil filters on the oil filter head.

Mechanical overtightening can distort the threads or damage the oil filter element seal.

Turn the oil filter until the seal contacts the oil filter head.

Turn an additional $\frac{3}{4}$ to 1 turn.



Fill

Install the oil drain plugs and new copper washers into the oil sump and oil pan adapter cover plate.

Tighten the oil drain plugs.

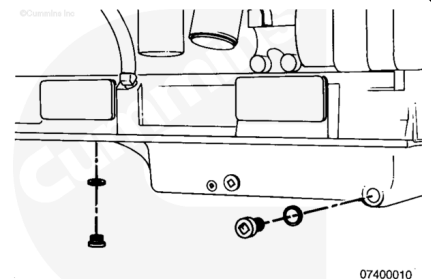
Torque Value: 100 n•m [74 ft-lb]

Fill the engine with clean 15W-40 oil. For oil pan capacities,

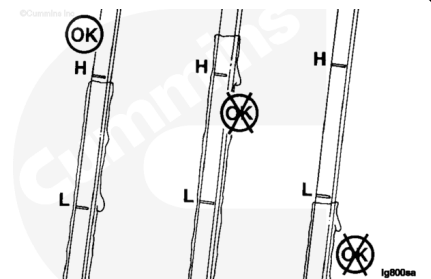
Refer to Procedure 018-017

(</qs3/pubsys2/xml/en/procedures/18/18-018-017.html>)

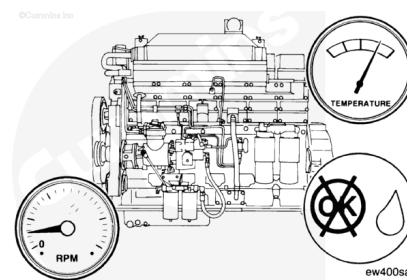
(Lubricating Oil System Specifications) in Section V.



Check the oil level on the dipstick. The level **must** be to the "H" (high) mark on the dipstick.



Operate the engine to normal operating temperature and check for oil leaks.



Shut the engine off and wait five to seven minutes to allow the oil to drain back into the oil pan.

Check the oil level.

Add oil as necessary to bring the level up to the "H" (high) mark on the dipstick.

